

StatusL2 Audit Summary

Prepared by Cyfrin

Version 2.0

Lead Auditors

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for StatusL2 on StatusL2. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated January 5, 2026, was delivered to StatusL2, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 15 days, the Cyfrin team conducted an audit on the StatusL2 code provided by StatusL2. The findings consist of 2 Critical, 6 High, 8 Medium & 8 Low severity issues with the remainder being informational.

Summary

Project Name	StatusL2
Repository	status-network-monorepo
Commit	6ab031b9d97d...
Audit Timeline	Dec 1st - Dec 19th, 2025
Methods	Manual Review

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	2	2	0
High Risk	6	6	0
Medium Risk	8	7	1
Low Risk	8	8	0
Informational	9	7	2
Gas Optimizations	0	0	0
Total	33	30	3

6 Audit Scope

The audit scope was limited to:

```
// core
status-network-contracts/src/math/MultiplierPointMath.sol
status-network-contracts/src/math/StakeMath.sol
status-network-contracts/src/nft-metadata-generators/BaseNFTMetadataGenerator.sol
status-network-contracts/src/nft-metadata-generators/NFTMetadataGeneratorSVG.sol
status-network-contracts/src/nft-metadata-generators/NFTMetadataGeneratorURL.sol
status-network-contracts/src/rln/RLN.sol
status-network-contracts/src/rln/PoseidonHasher.sol
status-network-contracts/src/utils/ERC20VotesUpgradeable.sol
status-network-contracts/src/Karma.sol
status-network-contracts/src/KarmaAirdrop.sol
status-network-contracts/src/KarmaNFT.sol
status-network-contracts/src/KarmaTiers.sol
status-network-contracts/src/StakeManager.sol
status-network-contracts/src/SimpleKarmaDistributor.sol
status-network-contracts/src/StakeVault.sol
status-network-contracts/src/TrustedCodehashAccess.sol
status-network-contracts/src/VaultFactory.sol

// deployment scripts
status-network-contracts/script/Base.s.sol
status-network-contracts/script/DeployKarma.s.sol
status-network-contracts/script/DeployKarmaAirdrop.s.sol
status-network-contracts/script/DeployKarmaNFT.s.sol
status-network-contracts/script/DeployKarmaTiers.s.sol
status-network-contracts/script/DeploymentConfig.s.sol
status-network-contracts/script/DeployMetadataGenerator.s.sol
status-network-contracts/script/DeployProtocol.s.sol
status-network-contracts/script/DeploySimpleKarmaDistributor.s.sol
status-network-contracts/script/DeployStakeManager.s.sol
status-network-contracts/script/DeployVaultFactory.s.sol
status-network-contracts/script/RLN.s.sol
status-network-contracts/script/UpgradeKarma.s.sol
status-network-contracts/script/UpgradeStakeManager.s.sol
```

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

The StatusL2 KARMA protocol is a staking and reputation system designed to reward long-term participation through time-based accrual mechanics. Users stake tokens into protocol-managed vaults and earn non-transferable Multiplier Points (MP) proportional to stake size and duration, subject to caps.

Stakes accrue MP linearly over time, with a per-stake maximum corresponding to a fixed accrual period (typically four years). Accrued MP contributes to a user's overall KARMA score and is used to determine tiers, reputation and reward eligibility.

The system is composed of staking vaults deployed via a factory pattern, with core logic implemented in StakeManager, StakeVault and supporting math libraries. MP balances feed into tiering and reputation modules (Karma, KarmaTiers) and are used by reward distributors to compute user entitlements.

Additional components include an airdrop mechanism based on Merkle proofs and signature delegation, and NFT representations of user KARMA state.

9 Attestation

Cyfrin attests that it performed a security review of the StatusL2 code provided by StatusL2.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2026-01-05-cyfrin-statusl2-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to StatusL2 has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=7de2ec1ad2fad63b4b7076f927c3a239fe2a4d12ea4b6de31bc8484e6a1fde0c
F=2026-01-05-cyfrin-statusl2-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.